Double Salts vs Coordination Complexes & Werner's Postulates (NCERT §9.1)

Coordination compounds are addition compounds containing a central metal atom or ion surrounded by an array of bound ions or neutral molecules (ligands). They retain their structural identity in solid state as well as in aqueous solution.

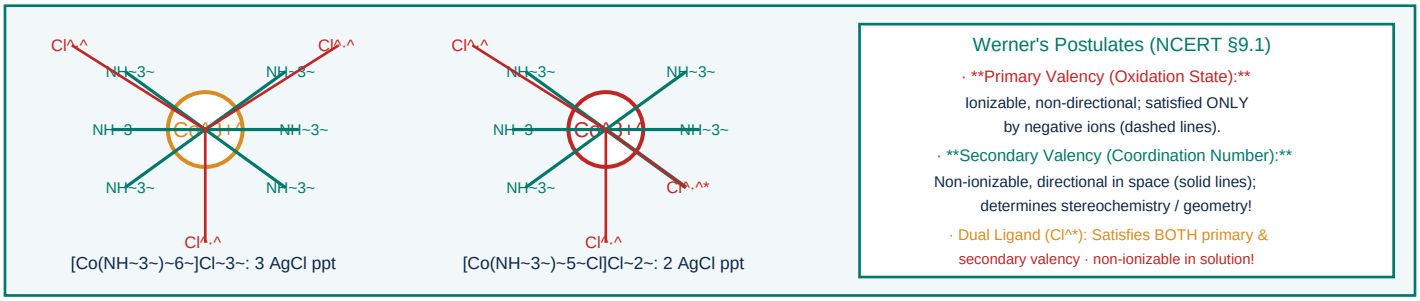


Fig. 1.1 — Primary (dashed, ionizable) vs Secondary (solid, directional) valencies in cobalt ammines.

1. Double Salts vs Coordination (Complex) Compounds

Property	Double Salts (e.g. Mohr's Salt, Alum, Carnallite)	Coordination Complexes (e.g. $K_4[Fe(CN)_6]$)
Solution Behavior	Completely dissociate into constituent simple ions in water	Retain identity; complex ion does NOT dissociate into simple ions
Chemical Tests	Give positive confirmatory tests for ALL individual constituent ions	Do not give tests for constituent ions locked inside coordination sphere
Valency Nature	Governed strictly by normal electrovalent / ionic bonds	Involves coordinate covalent (dative) bonds from ligands to metal
Classic Examples	Mohr's Salt: $\text{FeSO}_4\cdot(\text{NH}_4)_2\text{SO}_4\cdot 6\text{H}_2\text{O}$; Potash Alum: $\text{K}_2\text{SO}_4\cdot\text{Al}_2(\text{SO}_4)_3\cdot 24\text{H}_2\text{O}$	Potassium Ferrocyanide: $\text{K}_4[\text{Fe(CN)}_6] \rightarrow 4\text{K}^+ + [\text{Fe(CN)}_6]^{4-}$ (No Fe^{2+}!)

Table 1.1 — Diagnostic comparison between double salts and coordination complexes.

2. Alfred Werner's Coordination Theory Postulates

- **1. Two Types of Valencies:** Transition metals exhibit two distinct types of valency: **Primary Valency** and **Secondary Valency**.
- **2. Primary Valency (Ionizable, Non-directional):** Corresponds to the **formal oxidation state** of the central metal. Satisfied exclusively by negative ions (anions). In structural formulas, represented by **dashed lines** (---).
- **3. Secondary Valency (Non-ionizable, Directional):** Corresponds to the **Coordination Number (CN)** of the central metal. Satisfied by neutral molecules or anions. Directed towards fixed positions in space, dictating the **three-dimensional stereochemistry / geometry** (e.g. octahedral, tetrahedral, square planar). In structural formulas, represented by **solid lines** (—).
- **4. Dual Behavior of Ligands:** An anion may simultaneously satisfy BOTH primary and secondary valencies (e.g. $[\text{Co(NH}_3)_5\text{Cl}]\text{Cl}_2$, where one Cl^- is inside the coordination sphere). It does not ionize in solution!

TRAP
CRITICAL TRAP: When $[\text{Co(NH}_3)_4\text{Cl}_2]\text{Cl}$ is treated with excess AgNO_3, exactly **1 mole of AgCl** precipitates per mole of complex, because only the outer-sphere chloride ion is ionizable! The two inner chlorides satisfy secondary valency.

Quantitative Experimental Verification of Werner's Formulations (NCERT Table 9.1)

Alfred Werner established the correct structural formulas of cobalt(III) ammine complexes by correlating electrical molar conductivity, cryoscopic depression of freezing point (van 't Hoff factor i), and quantitative precipitation of chloride ions with excess ext{AgNO}_3.

1. Analysis of the Classic Cobalt(III) Ammine Series

Traditional Name / Color	Empirical Formula	Modern Werner Formulation	Molar Conductivity / Electrolyte Type	Moles of AgCl ppt per mole of complex
Luteo complex (Yellow)	\text{CoCl}_3\cdot 6\text{NH}_3	[\text{Co(NH}_3)_6]\text{Cl}_3	**1:3 electrolyte** (4 ions total: [\text{Co(NH}_3)_6]^{3+} + 3\text{Cl}^-)	**3 moles AgCl**
Purpureo complex (Purple)	\text{CoCl}_3\cdot 5\text{NH}_3	[\text{Co(NH}_3)_5\text{Cl}]\text{Cl}_2	**1:2 electrolyte** (3 ions total: [\text{Co(NH}_3)_5\text{Cl}]^{2+} + 2\text{Cl}^-)	**2 moles AgCl**
Praseo complex (Green, trans)	\text{CoCl}_3\cdot 4\text{NH}_3	[\text{Co(NH}_3)_4\text{Cl}_2]\text{Cl}	**1:1 electrolyte** (2 ions total: [\text{Co(NH}_3)_4\text{Cl}_2]^+ + \text{Cl}^-)	**1 mole AgCl**
Violeo complex (Violet, cis)	\text{CoCl}_3\cdot 4\text{NH}_3	[\text{Co(NH}_3)_4\text{Cl}_2]\text{Cl}	**1:1 electrolyte** (2 ions total: [\text{Co(NH}_3)_4\text{Cl}_2]^+ + \text{Cl}^-)	**1 mole AgCl**
Neutral complex	\text{CoCl}_3\cdot 3\text{NH}_3	[\text{Co(NH}_3)_3\text{Cl}_3]	**Non-electrolyte** (0 ions, non-conducting solution)	**0 moles AgCl** (No ppt!)

Table 2.1 — Werner's cobalt ammine series: formulation, ions, and AgCl stoichiometry.

2. Physical Measurements Confirming Werner's Formulations

- **1. Molar Electrical Conductivity (Λ_m):** Directly proportional to the number of free ions produced in solution: \Lambda_m(\text{1:3}) > \Lambda_m(\text{1:2}) > \Lambda_m(\text{1:1}) > \text{Non-electrolyte} \approx 0.
- **2. Colligative Properties (van 't Hoff factor i):** Depression of freezing point (\Delta T_f = i K_f m) confirms the total particle count: i = 4 for [ext{Co(NH}_3)_6] ext{Cl}_3; i = 3 for [ext{Co(NH}_3)_5 ext{Cl}] ext{Cl}_2; i = 2 for [ext{Co(NH}_3)_4 ext{Cl}_2] ext{Cl}; i = 1 for [ext{Co(NH}_3)_3 ext{Cl}_3].
- **3. Reagent Test with H_2SO_4:** In [ext{Co(NH}_3)_5 ext{SO}_4] ext{Br}, adding ext{BaCl}_2 produces NO white precipitate of ext{BaSO}_4 because sulfate is inside the coordination sphere! Adding ext{AgNO}_3 produces pale yellow ext{AgBr}\downarrow instantly.

TRAP

JEE ADVANCED TRAP: When 100\text{ mL} of 0.1\text{ M } [\text{Pt(NH}_3)_4\text{Cl}_2]\text{Cl}_2 is treated with excess ext{AgNO}_3, millimoles of AgCl precipitated = 100 \times 0.1 \times 2 = 20\text{ mmol} (0.02 mol), NOT 40 mmol!

Anatomy of a Coordination Entity (NCERT §9.2)

A coordination compound consists of a central coordination entity surrounded by ionizable counter ions. Mastery of coordination definitions is essential for IUPAC naming, isomerism, and crystal field analysis.

1. Structural Components of Coordination Entities

- **Central Atom / Ion:** The metal atom or cation to which a fixed number of neutral molecules or anions are bound in a definite geometrical arrangement. Acts as a **Lewis Acid** (electron pair acceptor).
- **Ligands:** The neutral molecules, anions, or cations bound to the central metal atom/ion by donating at least one electron pair. Act as **Lewis Bases** (electron pair donors).
- **Coordination Sphere:** The central atom/ion together with all its directly attached ligands enclosed within **square brackets** [\dots]. It behaves as a single discrete, non-ionizable chemical unit in solution.
- **Ionization Sphere (Counter Ions):** The ionizable species written outside the square brackets that balance the net electrical charge of the coordination entity (e.g. K^+ in $\text{K}_4[\text{Fe(CN)}_6]$, Cl^- in $[\text{Co(NH}_3)_6]\text{Cl}_3$).

2. Coordination Number (CN) & Coordination Polyhedra

The **Coordination Number (CN)** of a central metal ion in a complex is defined as the **total number of ligand donor atoms** to which the metal is directly bonded: $\mathbf{\text{CN}} = \sum (\text{Number of ligands} \times \text{Denticity})$ • For monodentate ligands: $\text{CN} = \text{number of ligands}$. E.g. in $[\text{PtCl}_6]^{2-}$, $\text{CN} = 6$. • For bidentate ligands: each ligand contributes 2 donor bonds. E.g. in $[\text{Fe(C}_2\text{O}_4)_3]^{3-}$, $\text{CN} = 3 \times 2 = \mathbf{6}$! In $[\text{Co(en)}_3]^{3+}$, $\text{CN} = 3 \times 2 = \mathbf{6}$! • In $[\text{Fe(EDTA)}]^-$, EDTA^{4-} is hexadentate $\implies \mathbf{\text{CN} = 6}$.

3. Calculation of Central Metal Oxidation State (OS)

Complex Species	Ligand Charges	Algebraic Charge Equation	Calculated Metal OS
$[\text{Fe(CN)}_6]^{4-}$	$\text{CN}^- = -1$	$x + 6(-1) = -4 \implies x = +2$	**Fe(+II)**
$[\text{Co(NH}_3)_5\text{Cl}]^{2+}$	$\text{NH}_3 = 0$, $\text{Cl}^- = -1$	$x + 5(0) + (-1) = +2 \implies x = +3$	**Co(+III)**
$[\text{Ni(CO)}_4]$	$\text{CO} = 0$	$x + 4(0) = 0 \implies x = 0$	**Ni(0)** (Zero valent!)
$[\text{Fe(H}_2\text{O)}_5(\text{NO})]\text{SO}_4$	$\text{H}_2\text{O} = 0$, $\text{NO}^+ = +1$, $\text{SO}_4^{2-} = -2$	$x + 5(0) + (+1) - 2 = 0 \implies x = +1$	**Fe(+I)** (Brown ring complex)
$[\text{Pt(NH}_3)_2\text{Cl(NO}_2)]$	$\text{NH}_3 = 0$, $\text{Cl}^- = -1$, $\text{NO}_2^- = -1$	$x + 2(0) - 1 - 1 = 0 \implies x = +2$	**Pt(+II)**

Table 3.1 — Oxidation state calculations of representative coordination complexes.

TRAP

EXAM TRAP: Do not confuse Coordination Number with Oxidation State! In $[\text{Fe(C}_2\text{O}_4)_3]^{3-}$, the Coordination Number is 6, while the Oxidation State of iron is +3.

Complete Classification of Ligands by Electron Donation & Topology (NCERT §9.2)

Ligands are categorized by charge (anionic, neutral, cationic) and by **denticity** (the number of donor atoms through which a single ligand binds simultaneously to the central metal atom).

1. Monodentate, Bidentate & Polydentate Ligands

Denticity Class	Ligand Name & Abbreviation	Donor Atoms	Charge	Coordination Notes
Monodentate	Aqua (H_2O), Ammine (NH_3)	O, N	0	Neutral molecules with lone pairs
Monodentate	Halido (F^-, Cl^-, Br^-, I^-)	Halogen	−1	Typical monoatomic anionic ligands
Bidentate	Ethylenediamine (**en**)	**2 N atoms**	0	Neutral symmetrical chelate; $\text{H}_2\text{N-CH}_2\text{-CH}_2\te
Bidentate	Oxalato (**ox**)	**2 O atoms**	−2	Anionic symmetrical chelate; $[\text{C}_2\text{O}_4]^{2-}$
Bidentate	Glycinato (**gly**)	**1 N, 1 O**	−1	Unsymmetrical bidentate ligand; $\text{H}_2\text{N-CH}_2\text{-COO}^-$
Bidentate	Dimethylglyoximato (**dmg**)	**2 N atoms**	−1	Forms scarlet red $[\text{Ni(dmg)}_2]$ stabilized by H-bonding
Polydentate	Diethylenetriamine (**dien**)	**3 N atoms**	0	Tridentate neutral chelating ligand
Polydentate	Triethylenetetramine (**trien**)	**4 N atoms**	0	Tetradentate neutral chelating ligand
Hexadentate	Ethylenediaminetetraacetate (**EDTA**)	**2 N + 4 O**	**−4**	Hexadentate master chelate; traps Ca^{2+}, Mg^{2+}, Pb^{2+}

Table 4.1 — Comprehensive classification of ligands by denticity and donor atoms.

2. Ambidentate Ligands (Linkage Isomerism Source)

A monodentate ligand possessing **two distinct donor atoms**, but able to bind through **ONLY ONE donor atom at a time**, is called an **ambidentate ligand**: · **Nitrito-N vs Nitrito-O:** $\text{M} \leftarrow \mathbf{\text{NO}_2^-}$ (Nitro / Nitrito-N, binds via N) vs $\text{M} \leftarrow \mathbf{\text{ONO}^-}$ (Nitrito / Nitrito-O, binds via O). · **Thiocyanato vs Isothiocyanato:** $\text{M} \leftarrow \mathbf{\text{SCN}^-}$ (Thiocyanato-S, binds via S) vs $\text{M} \leftarrow \mathbf{\text{NCS}^-}$ (Isothiocyanato-N, binds via N). · **Cyanido vs Isocyanido:** $\text{M} \leftarrow \mathbf{\text{CN}^-}$ (Cyanido-C) vs $\text{M} \leftarrow \mathbf{\text{NC}^-}$ (Cyanido-N / Isocyanido).

3. Flexidentate Ligands

Polydentate ligands that do not always utilize their maximum denticity, but adjust their number of bound donor atoms depending on the metal and reaction conditions: · **EDTA^{4-}:** Hexadentate in $[\text{Co(EDTA)}]^-$, but acts as **pentadentate** in $[\text{Co(EDTA)Br}]^{2-}$ where one carboxylate group remains uncoordinated! · **Sulfate (SO_4^{2-}):** Usually monodentate, but can act as bidentate chelate or bridging ligand in binuclear complexes.

TRAP
CRITICAL DISTINCTION: Polydentate ligands donate through multiple donor atoms SIMULTANEOUSLY forming rings (chelates). Ambidentate ligands possess two possible donor atoms but coordinate through ONLY ONE at a given instant!

Thermodynamic Driving Force of Ring-Forming Complexes (NCERT §9.8)

When a di- or polydentate ligand binds to a central metal ion to form a **5- or 6-membered heterocyclic ring**, the complex is called a **chelate**, and the phenomenon is known as the **chelate effect**.

1. The Thermodynamic Origin of the Chelate Effect

Chelated complexes are exponentially more thermodynamically stable than analogous non-chelated complexes containing monodentate ligands with the same donor atoms: $[\text{Ni(H}_2\text{O)}_6]^{2+} + 6\text{NH}_3 \rightleftharpoons [\text{Ni(NH}_3)_6]^{2+} + 6\text{H}_2\text{O} \quad (\log \beta = 8.61)$ $[\text{Ni(H}_2\text{O)}_6]^{2+} + 3\text{en} \rightleftharpoons [\text{Ni(en)}_3]^{2+} + 6\text{H}_2\text{O} \quad (\log \beta = 18.28)$ The complex with ethylenediamine is nearly 10^{10} times more stable!

- **Entropy Driven Phenomenon ($\Delta S > 0$):** In the reaction with en, 4 reactant particles (1 \text{ complex} + 3 \text{ en}) produce **7 product particles** (1 \text{ complex} + 6 \text{ free } \text{H}_2\text{O}).
- **Thermodynamic Consequence:** Release of 6 ordered coordinated water molecules into bulk solvent generates a massive increase in translational entropy: **ΔS° is large and POSITIVE (+ve)**! Since $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$, this makes ΔG° heavily negative.
- **Ring Size Rule:** 5-membered chelate rings are most stable with saturated aliphatic diamines (en); 6-membered rings are most stable when conjugated resonance is present (acetylacetonate, **acac**).

2. The Macrocyclic Effect

Complexes of **macrocyclic polydentate ligands** (e.g. crown ethers, cryptands, porphyrin ring in heme/chlorophyll) are significantly more stable than complexes of open-chain polydentate ligands with the same donor atoms. This enhanced stability arises because macrocyclic ligands are **pre-organized in space** (smaller loss of conformational entropy upon binding).

3. Practical Applications of Chelation

- **1. Water Hardness Estimation:** Titration of hard water with $\text{Na}_2\text{H}_2\text{EDTA}$ using Eriochrome Black-T (EBT) indicator selectively chelates Ca^{2+} and Mg^{2+}.
- **2. Chelation Therapy (Antidotes):** EDTA^{4-} is used to treat **lead poisoning** by forming stable water-soluble $[\text{Pb(EDTA)}]^{2-}$ excreted in urine. **D-penicillamine** chelates excess copper in Wilson's disease. **Desferrioxamine B** removes toxic iron overload.

TRAP

EXAM TRAP: Dimethylglyoxime (dmg^-) forms an extraordinarily stable bis-complex with Ni^{2+}: $[\text{Ni(dmg)}_2]$. Its stability is bolstered by TWO intramolecular **O-H···O hydrogen bonds** holding the two planar rings in a rigid square planar geometry!

Systematic IUPAC Nomenclature Rules & Formula Standards (NCERT §9.3)

IUPAC nomenclature provides unambiguous rules for naming neutral, cationic, and anionic coordination complexes, bridge ligands, and structural isomers.

1. Systematic Rules for Naming Coordination Compounds

- **1. Order of Cation and Anion:** The cation is ALWAYS named first, followed by the anion (whether complex or simple). E.g. in \text{K}_4[\text{Fe(CN)}_6], Potassium is named first; in [\text{Co(NH}_3)_6]\text{Cl}_3, Hexaamminecobalt(III) is named first.
- **2. Inside the Coordination Entity:** Ligands are listed in **alphabetical order** based on their first letter, regardless of their charge or number, followed by the central metal name.
- **3. Ligand Name Modifications:**
- - Anionic ligands ending in -ide, -ite, -ate are changed to end in **-ido**, **-ito**, **-ato** (e.g. \text{Cl}^- \rightarrow chlorido, \text{CN}^- \rightarrow cyanido, \text{OH}^- \rightarrow hydroxido, \text{C}_2\text{O}_4^{2-} \rightarrow oxalato, \text{SO}_4^{2-} \rightarrow sulfato).
- - Neutral ligands retain regular names, with four key exceptions: \mathbf{\text{H}_2\text{O} = \text{aqua}}, \mathbf{\text{NH}_3 = \text{ammine}} (double 'm'!), \mathbf{\text{CO} = \text{carbonyl}}, \mathbf{\text{NO} = \text{nitrosyl}}.
- **4. Numerical Prefixes:** di-, tri-, tetra-, penta-, hexa-. When the ligand name already contains a numerical prefix (e.g. ethylenediamine) or is polydentate, use **bis-**, **tris-**, **tetrakis-** with the ligand enclosed in parentheses.
- **5. Metal Ending:**
- - In **cationic or neutral complexes**, the metal retains its standard English name (e.g. cobalt, platinum, iron).
- - In **anionic complexes**, the metal name MUST end in the suffix **-ate** (e.g. cobaltate, platinate, ferrate, cuprate, argentate, aurate).
- **6. Oxidation State:** Roman numeral enclosed in parentheses immediately following the metal name without space (e.g. iron(II), nickel(0)).

2. Master Worked IUPAC Examples

Chemical Formula	Ion Type	Correct IUPAC Name	High-Yield Error to Avoid
[\text{Co(NH}_3)_6]\text{Cl}_3	Cationic complex	**Hexaamminecobalt(III) chloride**	Do not write 'trichloride' for counter ion!
\text{K}_4[\text{Fe(CN)}_6]	Anionic complex	**Potassium hexacyanidoferrate(II)**	Must use Latin root 'ferrate', not 'ironate'!
\text{K}_2[\text{PtCl}_6]	Anionic complex	**Potassium hexachloridoplatinate(IV)**	Metal suffix must be '-ate'
[\text{Pt(NH}_3)_2\text{Cl(NO}_2)]	Neutral complex	**Diamminechloridonitrito-N-platinum(II)**	Alphabetical: ammine (a) < chlorido (c) < nitrito (n)
[\text{Co(en)}_3]_2(\text{SO}_4)_3	Cationic complex	**Tris(ethane-1,2-diamine)cobalt(III) sulfate**	Must use 'tris', not 'tri' for en!
[\text{Cr(en)}_2\text{Cl}_2]\text{Cl}	Cationic complex	**Dichloridobis(ethane-1,2-diamine)chromium(III) chloride**	Chlorido (c) precedes ethane-1,2-diamine (e)

Table 6.1 — Authoritative IUPAC names of high-yield JEE complexes.

TRAP

CRITICAL TRAP: For coordination isomers where both cation and anion are complexes, like [\text{Pt(NH}_3)_4][\text{PtCl}_4]: Name is **Tetraammineplatinum(II) tetrachloridoplatinate(II)**. The cationic platinum has NO '-ate' suffix, while the anionic platinum MUST end in '-ate'!

Constitutional Isomerism in Coordination Entities (NCERT §9.4.1)

Structural isomers possess the same molecular formula but differ in the connectivity of chemical bonds or distribution of ions between the inner coordination sphere and the outer ionization sphere.

1. The Four Primary Classes of Structural Isomerism

- **1. Ionization Isomerism:** *Arises when an ionizable counter-ion swaps places with a coordinated ligand within the coordination sphere, giving different ions in solution:*
- *- Pair A:* $[\text{Co(NH}_3)_5\mathbf{\text{SO}_4}]\mathbf{\text{Br}}$ *(Red-violet)* $\rightarrow$ *gives* Br^- *ion (pale yellow ppt with* AgNO_3*; no ppt with* BaCl_2*).*
- *- Pair B:* $[\text{Co(NH}_3)_5\mathbf{\text{Br}}]\mathbf{\text{SO}_4}$ *(Red)* $\rightarrow$ *gives* SO_4^{2-} *ion (white ppt of* BaSO_4 *with* BaCl_2*; no ppt with* AgNO_3*).*
- **2. Solvate / Hydrate Isomerism:** *Special case of ionization isomerism where water molecules differ in being coordinated as ligands vs trapped as crystal lattice solvent:*
- *-* $[\text{Cr(H}_2\text{O)}_6]\text{Cl}_3$ ***(Violet)****: 3 free* Cl^- *ions; 3 moles AgCl ppt; 0 lattice* H_2O *lost over* H_2SO_4*.*
- *-* $[\text{Cr(H}_2\text{O)}_5\text{Cl}]\text{Cl}_2\cdot\text{H}_2\text{O}$ ***(Blue-green)****: 2 free* Cl^- *ions; 2 moles AgCl ppt; 1 mole* H_2O *lost.*
- *-* $[\text{Cr(H}_2\text{O)}_4\text{Cl}_2]\text{Cl}\cdot 2\text{H}_2\text{O}$ ***(Dark green)****: 1 free* Cl^- *ion; 1 mole AgCl ppt; 2 moles* H_2O *lost.*
- **3. Linkage Isomerism:** *Occurs when an* ***ambidentate ligand*** *can bind to the metal through either of two different donor atoms:*
- *-* $[\text{Co(NH}_3)_5(\mathbf{\text{NO}_2})]\text{Cl}_2$ ***(Yellow-brown****, Pentaamminenitrito-N-cobalt(III) chloride; M-N bond).*
- *-* $[\text{Co(NH}_3)_5(\mathbf{\text{ONO}})]\text{Cl}_2$ ***(Red****, Pentaamminenitrito-O-cobalt(III) chloride; M-O bond).*
- **4. Coordination Isomerism:** *Occurs in binuclear compounds where both cation and anion are complex ions, resulting from mutual exchange of ligands between the two metal centers:*
- *- E.g.* $[\text{Co(NH}_3)_6][\text{Cr(CN)}_6]$ *and* $[\text{Cr(NH}_3)_6][\text{Co(CN)}_6]$*.*

2. Advanced Structural Variants for JEE Advanced

- **Ligand Isomerism:** *Isomerism originating within the organic ligand itself. E.g. 1,2-diaminopropane (pn) vs 1,3-diaminopropane (tn).*
- **Coordination Position Isomerism:** *In polynuclear bridged complexes, an exchange of non-bridging ligands between the two metal centers:*
- *-* $[(\text{NH}_3)_4\text{Co}(\mu\text{-NH}_2)(\mu\text{-O}_2)\text{Co}(\text{NH}_3)_2\text{Cl}$ $^{2+}$ *vs* $[(\text{NH}_3)_3\text{ClCo}(\mu\text{-NH}_2)(\mu\text{-O}_2)\text{Co}(\text{NH}_3)_3\text{}$ $^{2+}$*.*

TRAP

EXAM TRAP: $[\text{Pt(NH}_3)_4][\text{PtCl}_4]$ *and* $[\text{Pt(NH}_3)_3\text{Cl}][\text{Pt(NH}_3)\text{Cl}_3]$ *are COORDINATION ISOMERS! Check the oxidation states carefully to ensure both complexes remain chemically viable (+2 on both Pt).*

Spatial Arrangement of Ligands Around Metal Centers (NCERT §9.4.2)

Stereoisomers have the same chemical formula and atom-to-atom bonding sequence, but differ in the three-dimensional orientation of ligands in space. Geometrical isomerism arises when ligands occupy adjacent (cis) or opposite (trans) coordination sites.

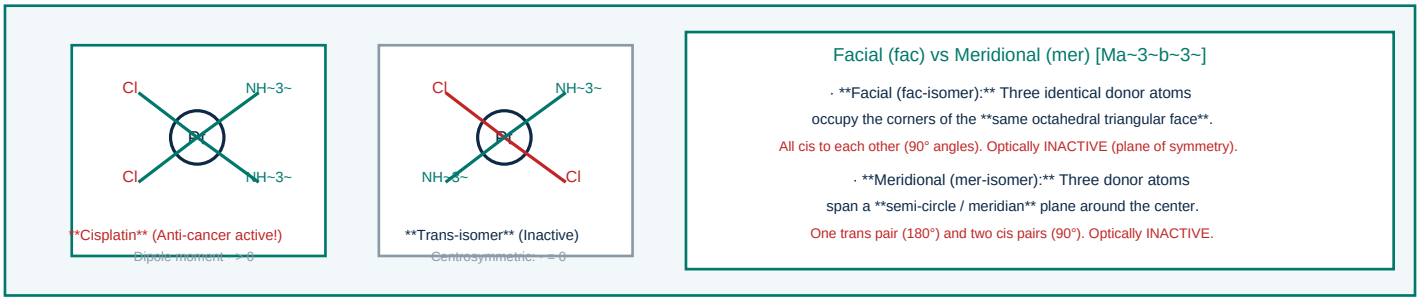


Fig. 8.1 — Cisplatin vs Trans-isomer and Facial (fac) vs Meridional (mer) octahedra.

1. Geometrical Isomerism in Coordination Number 4

- **Tetrahedral Complexes (CN = 4):** CANNOT exhibit geometrical isomerism because all four coordination positions are mutually adjacent and equidistant to each other (all bond angles are 109.5°)! E.g. [ext{Ma}_2 ext{b}_2], [ext{Mabcd}] have NO cis/trans isomers.
- **Square Planar Complexes (CN = 4, sp^2d or dsp^2):**
- - **$[Ma_2b_2]$:** Forms **2 geometrical isomers**: **Cis** (identical ligands adjacent at 90°) and **Trans** (identical ligands opposite at 180°). Classic: **Cisplatin** (cis-[\text{Pt(NH}_3)_2\text{Cl}_2], anti-cancer drug) vs **Transplatin** (inactive).
- - **$[Ma_2bc]$:** Forms **2 geometrical isomers** (cis and trans with respect to 'a').
- - **[Mabcd]:** Forms **3 geometrical isomers** (fix ligand 'a' and place 'b', 'c', 'd' trans to it).
- - **$[M(AB)_2]$:** Unsymmetrical bidentate ligand (e.g. [\text{Pt(gly)}_2]) forms **2 isomers** (cis and trans).

2. Geometrical Isomerism in Octahedral Complexes (CN = 6)

- **$[Ma_4b_2]$:** Forms **2 isomers** (Cis: 90° angle between 'b' ligands; Trans: 180° angle between 'b' ligands). E.g. [\text{Co(NH}_3)_4\text{Cl}_2]^+.
- **$[Ma_3b_3]$ — Facial (fac) and Meridional (mer) Isomerism:**
- - **Facial (fac-isomer):** Three donor atoms 'a' occupy the corners of a single triangular octahedral face. All three 'a' ligands are mutually cis (90°). E.g. fac-[\text{Co(NH}_3)_3(\text{NO}_2)_3].
- - **Meridional (mer-isomer):** Three donor atoms 'a' lie on a common plane (the meridian) passing through the metal center. Two 'a' ligands are trans (180°), while the third is cis (90°).
- **$[M(AA)_2b_2]$:** (e.g. [\text{Co(en)}_2\text{Cl}_2]^+) forms **Cis** and **Trans** isomers. The cis isomer lacks a center of inversion, while trans is centrosymmetric.

TRAP
CRITICAL TRAP: [ext{Ma}_6] and [ext{Ma}_5 ext{b}] CANNOT show geometrical isomerism in octahedral coordination! At least two non-identical ligands are required.

Chirality, Symmetry Elements & Isomer Counting Tables (NCERT §9.4.2)

Optical isomers (enantiomers) are non-superimposable mirror images that rotate the plane of polarized light in opposite directions (dextro '+' and laevo '–'). Chirality requires the complete **absence of improper rotation axes (no plane of symmetry** \sigma **and no center of inversion** i**)**.

1. Optical Activity in Octahedral Complexes

- **1. [M(AA)$_3$] (Tris-chelate complexes):** E.g. [\text{Co(en)}_3]$^{3+}$, [\text{Cr(ox)}_3]$^{3-}$. Belong to chiral point group D_3. Completely devoid of any plane of symmetry \sigma or center of inversion i \implies **Chiral and optically active!** Exist as non-superimposable \Delta (right-handed propeller) and \Lambda (left-handed propeller) enantiomeric pair.
- **2. [M(AA)$_2$b$_2$] (Bis-chelate complexes):** E.g. [\text{Co(en)}_2\text{Cl}_2]$^{+}$:
- - **Trans-isomer:** Possesses a plane of symmetry (and center of inversion) \implies **Optically INACTIVE (achiral)**.
- - **Cis-isomer:** Lacks any plane or center of symmetry \implies **Optically ACTIVE (chiral)**! Exists as d- and l-enantiomers.
- **3. [M(AA)b$_2$c$_2$]:** Cis-isomers are optically active; trans-isomers are optically inactive.
- **4. Square Planar Inactivity:** Square planar complexes possess a molecular plane of symmetry (the plane containing the metal and ligands) \implies **almost always optically INACTIVE** (rare exceptions occur with unsymmetrical bulky sterically hindered ligands).

2. Master Isomer Counting Table (JEE Advanced Core)

Complex Type	Geometrical Isomers (GI)	Optically Active Pairs (Enantiomers)	Total Stereoisomers (GI + Enantiomeric excess)
[\text{Ma}_4\text{b}_2]	2 (cis, trans)	0 (both achiral)	**2**
[\text{Ma}_3\text{b}_3]	2 (fac, mer)	0 (both achiral)	**2**
[\text{Ma}_2\text{b}_2\text{c}_2]	5	1 pair (all-cis is chiral)	**6** (5 GI + 1 enantiomer)
[\text{Mabcdef}]	**15** (all distinct!)	**15 pairs (all chiral!)**	**30** (15 pairs of enantiomers)
[\text{M(AA)}_3]	1	1 pair (chiral \Delta/\Lambda)	**2**
[\text{M(AA)}_2\text{b}_2]	2 (cis, trans)	1 pair (cis is chiral)	**3** (trans + cis-d + cis-l)
[\text{M(AA)b}_4]	1	0	**1**

Table 9.1 — Authoritative isomer counts for octahedral coordination complexes.

TRAP
EXAM TRAP: For [\text{Co(en)}_2\text{Cl}_2]$^{+}$, the number of GEOMETRICAL isomers is 2 (cis, trans), but the TOTAL number of STEREOISOMERS is 3 (trans + d-cis + l-cis)!

Effective Atomic Number & Orbital Hybridization Model (NCERT §9.5.1)

Valence Bond Theory (VBT), developed by Linus Pauling, describes coordination bonding as the overlap of empty hybrid orbitals of the central metal with filled lone-pair orbitals of ligands.

1. Sidgwick's Effective Atomic Number (EAN) Rule

Sidgwick proposed that complex stability is maximized when the central metal attains the electronic configuration of the next noble gas (Kr = 36, Xe = 54, Rn = 86): $\mathbf{\text{EAN}} = Z - (\text{Oxidation State}) + 2 \times (\text{Coordination Number})$ • In $\text{K}_4[\text{Fe(CN)}_6]$: $\text{EAN} = 26 - (+2) + 2(6) = 24 + 12 = \mathbf{36\text{ (Kr)}}$ $\implies$ Obeys EAN rule. • In $[\text{Ni(CO)}_4]$: $\text{EAN} = 28 - (0) + 2(4) = 28 + 8 = \mathbf{36\text{ (Kr)}}$ $\implies$ Obeys EAN rule. • In $[\text{Cu(NH}_3)_4]^{2+}$: $\text{EAN} = 29 - (+2) + 2(4) = 27 + 8 = \mathbf{35}$ $\implies$ Violates EAN rule (stable nonetheless).

2. Valence Bond Theory Postulates & Hybridization Types

- **1. Empty Hybrid Orbitals:** Metal ion provides empty hybrid orbitals matching its CN: sp^3 (Tetrahedral), dsp^2 (Square Planar), d^2sp^3 (Inner Octahedral), sp^3d^2 (Outer Octahedral).
- **2. Inner vs Outer Orbital Complexes:**
- - **Inner Orbital Complex (d^2sp^3):** Uses inner (n-1)d orbitals (3d). Requires pairing of unpaired d-electrons by strong field ligands $\implies$ typically **low spin / diamagnetic**.
- - **Outer Orbital Complex (sp^3d^2):** Uses outer nd orbitals (4d). Unpaired electrons remain unpaired in (n-1)d $\implies$ typically **high spin / paramagnetic**.

3. Five Standard VBT Worked Cases

Complex Ion	Config. / Metal OS	Hybridization	Geometry	Magnetic Property & Unpaired e^-
$[\text{Fe(CN)}_6]^{4-}$	Fe^{2+} (d^6), strong CN^- pairs 3d	d^2sp^3	Octahedral	**Diamagnetic** (n = 0, μ = 0)
$[\text{Fe(H}_2\text{O)}_6]^{2+}$	Fe^{2+} (d^6), weak H_2O no pairing	sp^3d^2	Octahedral	**Paramagnetic** (n = 4, μ = 4.90 BM)
$[\text{Co(NH}_3)_6]^{3+}$	Co^{3+} (d^6), strong NH_3 pairs 3d	d^2sp^3	Octahedral	**Diamagnetic** (n = 0, μ = 0)
$[\text{Ni(CN)}_4]^{2-}$	Ni^{2+} (d^8), strong CN^- pairs 3d	dsp^2	Square Planar	**Diamagnetic** (n = 0, μ = 0)
$[\text{NiCl}_4]^{2-}$	Ni^{2+} (d^8), weak Cl^- no pairing	sp^3	Tetrahedral	**Paramagnetic** (n = 2, μ = 2.83 BM)

Table 10.1 — Classical VBT cases: hybridization, geometry, and magnetic properties.

TRAP
CRITICAL VBT FLAW: VBT cannot explain why $[\text{Co(NH}_3)_6]^{3+}$ is diamagnetic with inner d^2sp^3 while $[\text{CoF}_6]^{3-}$ is paramagnetic with outer sp^3d^2! It cannot predict spectral absorption or color.

Electrostatic Crystal Field & Degeneracy Breakdown (NCERT §9.5.2)

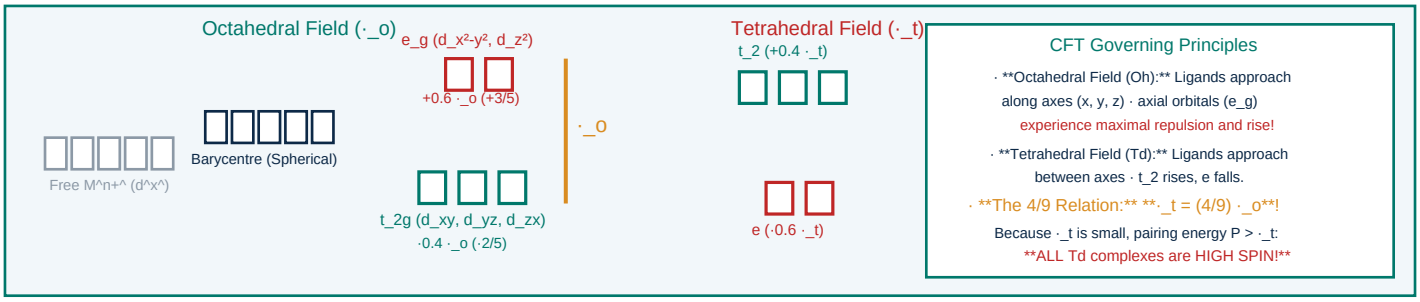


Fig. II.1 — Octahedral (Δ_o) and Tetrahedral (Δ_t) crystal field splitting diagrams.

1. Octahedral Splitting Mechanics & Pairing Criteria

- **Axial Splitting:** 6 ligands approach along axes. **e_g Set** ($d_{x^2-y^2}$, d_{z^2}) experiences direct axial repulsion \implies elevated by **+0.6** Δ_o. **t_2g Set** (d_{xy}, d_{yz}, d_{zx}) avoids ligands \implies stabilized by **−0.4** Δ_o.
- **CFSE Formula:** $\mathbf{\text{CFSE}_{oct}} = \left[-0.4\, n(t_{2g}) + 0.6\, n(e_g) \right] \Delta_o + mP$ (P = \text{pairing energy}).
- **High Spin vs Low Spin:** \Delta_o < P (*weak field*) \implies **High Spin** ($t_{2g}^3\ e_g^1$). \Delta_o > P (*strong field*) \implies **Low Spin** ($t_{2g}^4\ e_g^0$, *pairing favored*).

2. High-Spin vs Low-Spin Configurations & CFSE

Config.	High Spin (t_{2g}, e_g)	HS CFSE	HS Unp.	Low Spin (t_{2g}, e_g)	LS CFSE	LS Unp.
d^4	$t_{2g}^3\ e_g^1$	-0.6 \Delta_o	4	$t_{2g}^4\ e_g^0$	-1.6 \Delta_o + P	2
d^5	$t_{2g}^3\ e_g^2$	**0.0** (Zero)	5	$t_{2g}^5\ e_g^0$	-2.0 \Delta_o + 2P	1
d^6	$t_{2g}^4\ e_g^2$	-0.4 \Delta_o	4	$t_{2g}^6\ e_g^0$	**−2.4 Δ_o + 2P**	**0 (Diamag)**
d^7	$t_{2g}^5\ e_g^2$	-0.8 \Delta_o	3	$t_{2g}^6\ e_g^1$	-1.8 \Delta_o + P	1

Table II.1 — High-spin vs low-spin octahedral configurations and CFSE.

> *TRAP*
> *CRITICAL TRAP: For d^8, d^9, d^{10}, there is ONLY ONE configuration regardless of field: d^8 is always $t_{2g}^6\ e_g^2$ (n = 2), d^9 is $t_{2g}^6\ e_g^3$ (n = 1), and d^{10} is $t_{2g}^6\ e_g^4$ (n = 0).*

Inverted Splitting & Factors Dictating Crystal Field Magnitude (NCERT §9.5.2)

In tetrahedral complexes, four ligands approach from the corners of a cube containing the metal at its center, resulting in an inverted crystal field splitting pattern.

1. Tetrahedral Field Splitting (T_d Field)

- **Inverted Orbitals:** Ligands approach along cube corners (between axes), directly closer to d_{xy}, d_{yz}, d_{zx} than to d_{x^2-y^2}, d_{z^2}.
- - **t_2 Set (**d_{xy}, d_{yz}, d_{zx}**):** Elevated by **+0.4** Δ_t above the barycentre.
- - **e Set (**d_{x^2-y^2}, d_{z^2}**):** Stabilized by **−0.6** Δ_t below the barycentre.
- - Note: Subscript 'g' is omitted in tetrahedral complexes because a tetrahedron lacks an inversion center (i).
- **The Quantitative Relationship with Octahedral Field:**
- $\mathbf{\Delta_t = \frac{4}{9} \Delta_o} \approx 0.44 \Delta_o$
- Because tetrahedral geometry has only 4 ligands (vs 6 in Oh) and none point directly at d-orbitals.
- **Consequence:** Since \Delta_t is always much smaller than the pairing energy (\Delta_t < P), **pairing NEVER occurs in tetrahedral complexes! ALL tetrahedral complexes are HIGH SPIN!**

2. The Spectrochemical Series

Experimentally determined order of ligands based on the increasing magnitude of crystal field splitting (\Delta_o) they produce: $\mathbf{I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < bpy < phen < NO_2^- < PPh_3 < CN^- < CO}$ • **Weak Field Ligands (Halides, Oxygen donors):** Small \Delta_o, form high-spin complexes. • **Strong Field Ligands (Nitrogen donors, Carbon donors):** Large \Delta_o, form low-spin complexes; **CO is the strongest field ligand** due to synergic \pi-backbonding!

3. Four Factors Influencing the Magnitude of Δ

- **1. Oxidation State of Metal:** Higher formal charge pulls ligands closer \implies larger repulsion: \Delta_o(M^{3+}) > \Delta_o(M^{2+}) by ~50% (e.g. [\text{Co(H}_2\text{O)}_6]^{3+} is low-spin, but [\text{Co(H}_2\text{O)}_6]^{2+} is high-spin!).
- **2. Transition Series (3d vs 4d vs 5d):** \Delta_o increases down the group by 30-50%: **5d > 4d > 3d** (diffuse orbitals interact more strongly). **All 4d and 5d complexes are LOW SPIN**, even with weak-field ligands like \text{Cl}^-!
- **3. Geometry:** \Delta_{sp} (1.74 \Delta_o) > \Delta_o > \Delta_t (0.44 \Delta_o).
- **4. Nature of Ligand:** Spectrochemical position dictated by \sigma-donation and \pi-acceptor ability.

TRAP

HIGH-YIELD JEE TRAP: In [\text{Co(ox)}_3]^{3-}, oxalate is an oxygen-donor ligand, yet the complex is **diamagnetic and low spin**! Reason: \text{Co}^{3+} has high charge density, boosting \Delta_o above P!

Crystal Field Splitting in D_4h Geometry & The d⁸ Configuration (NCERT §9.5.3)

Square planar geometry (D_{4h}) is derived conceptually from an octahedral complex by progressively pulling the two trans ligands along the z-axis to infinity (tetragonal distortion to the square planar limit).

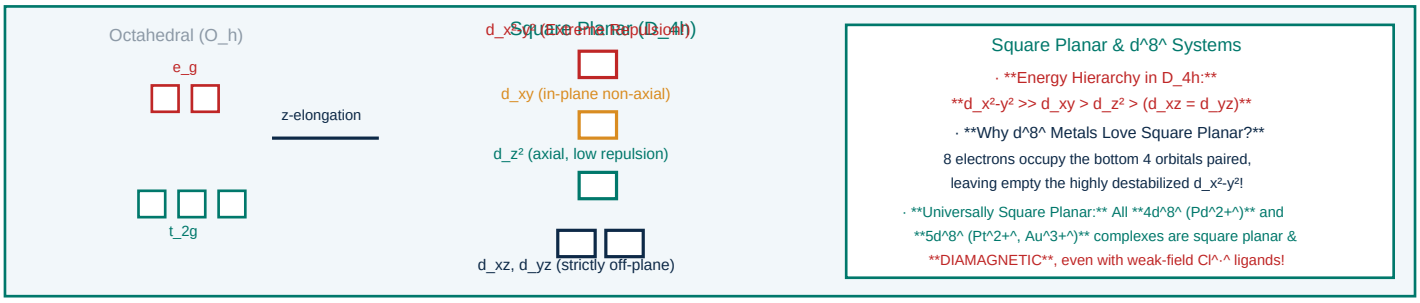


Fig. 13.1 — Octahedral to tetragonal to square planar (D_4h) crystal field energy levels.

1. Orbital Energy Splitting in D_4h Symmetry

- **1. $d_{x^2-y^2}$ (Highest Energy):** Lobes point directly at the 4 remaining in-plane ligands (x, y) \implies experiences maximal electrostatic repulsion; elevated to the highest energy level.
- **2. d_{xy} (Second Highest):** Lobes lie in the xy-plane between the ligands \implies substantial in-plane repulsion, but lower than $d_{x^2-y^2}$.
- **3. d_{z^2} (Third Highest):** Completely relieved of direct axial ligand encounters; stabilized substantially, lying below d_{xy} (its donut ring lies in the xy-plane).
- **4. d_{xz}, d_{yz} (Lowest Energy, Degenerate Pair):** Lobes point completely out of the coordination plane \implies experience minimum repulsion.
- **Master Energy Order in Square Planar:**
- $\mathbf{d_{x^2-y^2} \gg d_{xy} > d_{z^2} > (d_{xz} = d_{yz})}$
- The total square planar splitting energy is $\mathbf{\Delta_{sp} \approx 1.3 \text{ to } 1.74 \Delta_o}$.

2. Why d8 Metal Ions Universally Form Square Planar Complexes

In d^8 electronic systems (e.g., Ni^{2+}, Pd^{2+}, Pt^{2+}, Au^{3+}), the 8 d-electrons occupy the lower four orbitals in spin-paired configurations: $(d_{xz}, d_{yz})^4, (d_{z^2})^2, (d_{xy})^2$, leaving the strongly destabilized $d_{x^2-y^2}$ completely empty! · This produces an enormous Crystal Field Stabilization Energy (CFSE gain) that heavily favors square planar over tetrahedral geometry. · **All $4d^8$ (Pd^{2+}) and $5d^8$ (Pt^{2+}, Au^{3+}) complexes are SQUARE PLANAR and DIAMAGNETIC**, regardless of ligand field strength (even with weak Cl^- in $[\text{PtCl}_4]^{2-}$)!

3. Nickel(II) Complex Stereochemistry: [Ni(CN)4]2− vs [NiCl4]2−

Complex Ion	Ligand Type / Field	Geometry & Hybridization	Unpaired e^- (n)	Magnetic Character
$[\text{Ni(CN)}_4]^{2-}$	Strong field CN^-	**Square Planar (dsp^2)**	**0**	**Diamagnetic** (Yellow)
$[\text{NiCl}_4]^{2-}$	Weak field Cl^-	**Tetrahedral (sp^3)**	**2**	**Paramagnetic** ($\mu = 2.83 \text{ BM}$)
$[\text{Ni(CO)}_4]$	Strong field CO ($Ni^{0,\ 3d^{10}}$)	**Tetrahedral (sp^3)**	**0**	**Diamagnetic** (Colorless)

Table 13.1 — Contrasting geometries and magnetism in nickel complexes.

TRAP
EXAM TRAP: $[\text{PtCl}_4]^{2-}$ and $[\text{Pt(NH}_3)_4]^{2+}$ are BOTH square planar and diamagnetic because Pt is a 5d metal with huge Δ!

Electronic Degeneracy & Spontaneous Geometric Distortion (JEE Advanced Special)

The Jahn-Teller Theorem states: Any non-linear molecular system in a spatially degenerate electronic ground state is unstable and will undergo geometric distortion to lower its symmetry, remove the degeneracy, and lower the overall energy.

1. Asymmetric Electron Occupation: e_g vs t_2g

- **1. Strong Jahn-Teller Distortion (Asymmetric e_g Occupation):**
- Because e_g orbitals (d_{x^2-y^2}, d_{z^2}) point directly at ligands, an uneven electron distribution in e_g causes strong unequal repulsions on ligand pairs \implies **pronounced geometric distortion**.
- - **d^9 System (e.g. Cu^{2+}, $t_{2g}^6 e_g^3$):** Most famous case! One orbital has 2 electrons while the other has 1 electron.
- - **High-Spin d^4 System (e.g. Cr^{2+}, Mn^{3+}, $t_{2g}^3 e_g^1$):** Unequal occupation of e_g.
- - **Low-Spin d^7 System (e.g. Co^{2+} LS, Ni^{3+} LS, $t_{2g}^6 e_g^1$):** Pronounced distortion.
- **2. Weak Jahn-Teller Distortion (Asymmetric t_2g Occupation):**
- t_{2g} orbitals point between ligands \implies unequal electron distribution (d^1, d^2, \text{HS } d^6, \text{HS } d^7) produces only **very weak, subtle distortions** that are experimentally hard to observe.
- **3. Zero Jahn-Teller Distortion (Symmetrical Occupation):**
- d^3 (t_{2g}^3), HS d^5 ($t_{2g}^3 e_g^2$), LS d^6 (t_{2g}^6), d^8 ($t_{2g}^6 e_g^2$), and d^{10} ($t_{2g}^6 e_g^4$) have completely symmetrical sets \implies **zero distortion (perfect Oh symmetry)**!

2. Tetragonal Elongation (z-out) vs Compression (z-in)

In the vast majority of copper(II) complexes (e.g. [\text{Cu(H}_2\text{O)}_6]^{2+}): · If the 3rd e_g electron enters d_{z^2} ($d_{z^2}^2 d_{x^2-y^2}^1$), electrons shield the nucleus along the z-axis, weakening metal-ligand attraction \implies **two trans axial ligands move further away (z-elongation)**! · **Result:** 4 short in-plane equatorial bonds and **2 longer axial bonds** (e.g. in \text{CuF}_2, 4 equatorial Cu-F = 193 pm, 2 axial Cu-F = 227 pm!).

3. Copper(II) Ammonia Complex Stepwise Stability Drop

In [\text{Cu(NH}_3)_4(\text{H}_2\text{O)}_2]^{2+}, adding ammonia readily replaces the 4 in-plane equatorial water molecules to form [\text{Cu(NH}_3)_4]^{2+}, but the 5th and 6th water molecules on the elongated z-axis are so weakly bound that [\text{Cu(NH}_3)_6]^{2+} can ONLY be formed in anhydrous liquid ammonia!

TRAP

HIGH-YIELD JEE ADVANCED QUESTION: Which of the following shows strong Jahn-Teller distortion? (A) [\text{Cr(H}_2\text{O)}_6]^{3+} (d^3), (B) [\text{Mn(H}_2\text{O)}_6]^{2+} (d^5), (C) [\text{Cu(H}_2\text{O)}_6]^{2+} (d^9), (D) [\text{Fe(CN)}_6]^{4-} (d^6). **Answer is C** (d^9 asymmetric e_g)!

Electronic Transitions & Complementary Visible Absorption (NCERT §9.5.4)

The vivid colors of coordination complexes arise primarily from the absorption of specific wavelengths of visible light to excite electrons between crystal-field-split d-orbitals (d\text{-}d transitions).

1. Mechanism of Colour & The Complementary Wheel

- **Light Absorption:** When white light strikes a solution of a complex with an incompletely filled d-subshell (d^1 to d^9), a photon of energy $h\nu = \Delta_o$ is absorbed to promote an electron from t_{2g} to e_g.
- **Complementary Transmission:** The human eye perceives the light that is NOT absorbed — the **complementary color**: - **Absorbs Red** (long λ, small Δ) $\rightarrow$ Appears **Green**. - **Absorbs Yellow** $\rightarrow$ Appears **Violet / Blue**. - **Absorbs Blue-Green** $\rightarrow$ Appears **Red / Orange**. - **Absorbs Violet** (short λ, large Δ) $\rightarrow$ Appears **Yellow**.

2. Influence of Ligand on Color: [Ni(H2O)6]2+ Stepwise Substitution

Progressive replacement of weak H_2O by stronger ethylenediamine (en) increases Δ_o, shifting absorption to shorter wavelengths (higher frequency): 1. $[\text{Ni(H}_2\text{O)}_6]^{2+}$: **Green** (absorbs red light). 2. $[\text{Ni(H}_2\text{O)}_4(\text{en})]^{2+}$: **Pale blue**. 3. $[\text{Ni(H}_2\text{O)}_2(\text{en})_2]^{2+}$: **Blue / Purple**. 4. $[\text{Ni(en)}_3]^{2+}$: **Violet** (absorbs yellow light, largest Δ_o!).

3. Ruby and Emerald: Crystal Field Gemstones

- **Ruby:** Consists of colorless $\alpha\text{-Al}_2\text{O}_3$ in which $\sim 0.5\text{-}1\%$ of Al^{3+} sites are occupied by Cr^{3+} (d^3). In the tight alumina lattice, Al-O distance is short, compressing $\text{Cr}^{3+} \implies$ strong crystal field absorbs yellow-green light, transmitting **deep crimson red**!
- **Emerald:** Cr^{3+} replaces Al^{3+} in beryl ($\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$). Larger silicate cavities provide a slightly weaker crystal field, shifting absorption into the red/yellow region and transmitting **brilliant green**!

TRAP

WHY IS ANHYDROUS $CuSO_4$ WHITE? When white anhydrous CuSO_4 has no water ligands, the crystal field splitting is zero ($\Delta = 0$). All 5 d-orbitals remain degenerate; no $d\text{-}d$ transitions are possible $\implies$ pure white!

Organometallic Synergy: σ-Donation & π-Backbonding (NCERT §9.6)
Homoleptic metal carbonyls (e.g. [\text{Ni(CO)}_4], [\text{Fe(CO)}_5], [\text{Cr(CO)}_6]) exhibit unique metal–carbon bonding characterized by mutual reinforcement between \sigma and \pi bonds.

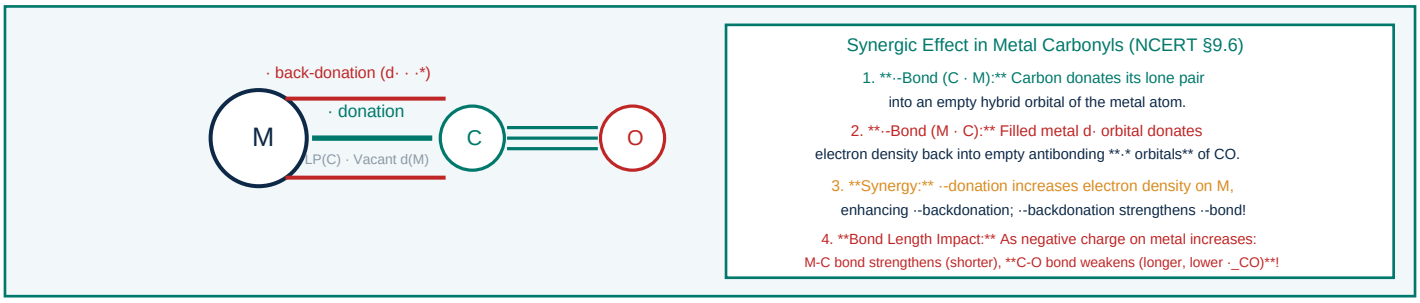


Fig. 16.1 — Synergic M–CO bonding: σ donation from C lone pair and π backbonding into CO π orbitals.

1. The Synergic Bonding Mechanism

- **1. M \leftarrow C \sigma-Bond:** *Carbon atom of CO donates its lone pair of electrons from a filled non-bonding sp-hybrid orbital into an empty hybrid orbital of the transition metal.*
- **2. M \rightarrow C \pi-Backbond:** *Filled d-orbitals (d_{xy}, d_{yz}, d_{zx}) of the transition metal overlap sideways with the* ***empty antibonding*** *\pi^* ***molecular orbitals*** *of the carbon monoxide ligand.*
- **3. The Synergic Loop:** *\sigma-donation increases electron density on the metal, facilitating \pi-backdonation; simultaneously, \pi-backdonation relieves excess negative charge from the metal, strengthening the \sigma-bond! This self-reinforcing cooperative loop is called* ***synergic bonding****.*

2. Impact on M–C and C–O Bond Orders & IR Stretching Frequencies (ν_CO)

When electrons are back-donated into the ***antibonding*** *\pi^* ***orbital*** *of CO: · Metal–Carbon (\text{M-C}) bond order increases (approaching double bond \text{M}=\text{C}): bond becomes* ***shorter and stronger****. · Carbon–Oxygen (\text{C-O}) bond order decreases (from triple bond \text{C}\equiv\text{O} towards double bond \text{C}=\text{O}): bond becomes* ***longer and weaker****! · By Hooke's Law (\nu = \frac{1}{2\pi c}\sqrt{k/\mu}), a weaker bond has a lower force constant k and thus a* ***lower IR stretching frequency*** *\nu_{\text{CO}}.*

3. Isoelectronic Carbonyl Series (JEE Advanced Master Metric)

Carbonyl Species	Metal & OS	Extrapolated π-Backdonation	M–C Bond Strength / Length	C–O Bond Order & IR \nu_{\text{CO}} (cm⁻¹)
[\text{Mn(CO)}_6]⁺	\text{Mn}(+1)	Least (positive charge holds d-electrons)	Weakest / Longest M–C	**Strongest C–O / Highest** \nu_{\text{CO}} (~2090 cm⁻¹)
[\text{Cr(CO)}_6]	\text{Cr}(0)	Intermediate	Intermediate M–C	Intermediate C–O (~2000 cm⁻¹)
[\text{V(CO)}_6]⁻	\text{V}(-1)	**Most (negative charge repels d-electrons)**	**Strongest / Shortest M–C**	**Weakest C–O / Lowest** \nu_{\text{CO}} (~1860 cm⁻¹)

Table 16.1 — Isoelectronic carbonyl comparison: charge vs backdonation and IR frequencies.

TRAP
CRITICAL JEE ADVANCED TRAP: *Higher negative charge on the metal carbonyl complex \implies GREATER \pi-backdonation \implies SHORTER M–C bond \implies LONGEST C–O bond \implies LOWEST \nu_{\text{CO}} stretching frequency!*

Stepwise Formation Constants & Stability Determinants (NCERT §9.8)

The stability of a complex in solution refers to the degree of association between the metal ion and the ligands involved in dynamic equilibrium.

1. Stepwise vs Overall Stability Constants (K_n vs β_n)

In solution, ligands add to a metal ion in a stepwise sequence: $\text{M} + \text{L} \rightleftharpoons \text{ML} \quad (K_1 = [\text{ML}] / [\text{M}][\text{L}]) \quad \text{ML} + \text{L} \rightleftharpoons \text{ML}_2 \quad (K_2 = [\text{ML}_2] / [\text{ML}][\text{L}]) \quad \text{Overall: } \text{M} + n\text{L} \rightleftharpoons \text{ML}_n \quad (\beta_n = K_1 \cdot K_2 \cdot \dots \cdot K_n)$ • In terms of logarithms: $\mathbf{\log \beta_n = \log K_1 + \log K_2 + \dots + \log K_n}$. · **Universal Trend:** Stepwise constants almost invariably follow: **$K_1 > K_2 > K_3 > \dots > K_n$** due to electrostatic repulsion, statistical probability of vacant sites, and steric crowding.

2. The Irving-Williams Series for Divalent 3d Metal Ions

For high-spin octahedral complexes of divalent 3d transition metal cations (M^{2+}), stability increases progressively regardless of the ligand nature, reaching a sharp maximum at Copper: $\mathbf{Ba^{2+} < Sr^{2+} < Ca^{2+} < Mg^{2+} < Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} > Zn^{2+}}$ • **Physical Driving Factors:** 1. **Ionic Size Decrease:** Radii contract from Mn^{2+} to Cu^{2+}, increasing electrostatic charge density (q/r). 2. **CFSE Gain:** Crystal field stabilization energy increases progressively from d^5 (Mn^{2+}, CFSE = 0) to d^8 (Ni^{2+}). 3. **The Copper Peak ($Cu^{2+} > Zn^{2+}$):** Copper(II) (d^9) experiences intense **Jahn-Teller distortion**, shortening 4 equatorial bonds and producing an exceptionally stable complex! Zn^{2+} (d^{10}, CFSE = 0) drops sharply in stability.

3. Thermodynamic vs Kinetic Stability

- **Thermodynamic Stability:** Measured by the equilibrium formation constant β_n (or $\Delta G^\circ = -RT \ln \beta_n$). High β indicates a stable complex.
- **Kinetic Stability (Lability vs Inertness):** Refers to the **rate of ligand substitution**: - **Labile Complexes:** Undergo ligand substitution very rapidly ($t_{1/2} < 1\text{ minute}$). E.g. $[\text{Cu(NH}_3)_4]^{2+}$ exchanges with $^{14}\text{NH}_3$ instantaneously. - **Inert Complexes:** Undergo ligand exchange very slowly ($t_{1/2} > 1\text{ minute}$). E.g. d^3 (Cr^{3+}) and low-spin d^6 (Co^{3+}) complexes are kinetically inert!

TRAP

CRITICAL TRAP: $[\text{Fe(CN)}_6]^{4-}$ is thermodynamically very stable (huge β), but kinetically labile. Conversely, $[\text{Co(NH}_3)_6]^{3+}$ is thermodynamically unstable in acid (breaks down to Co^{2+}), but kinetically so inert that it persists for days without decomposing!

Bio-Inorganic Role, Metallurgical Extraction & Medicine (NCERT §9.7)

1. Major Analytical, Metallurgical & Biological Applications

- **1. Analytical Chemistry:** Ni^{2+} reacts with dmg in ammoniacal solution to give a **scarlet red precipitate** $[\text{Ni(dmg)}_2]$ stabilized by O-H···O hydrogen bonds. Hardness of water is estimated with standard $\text{Na}_2\text{H}_2\text{EDTA}$ (EBT indicator).
- **2. Metallurgy & Metal Extraction: MacArthur-Forrest Cyanide Process:** Au/Ag leached with NaCN: $4\text{Au} + 8\text{CN}^- + 2\text{H}_2\text{O} + \text{O}_2 \rightarrow 4[\text{Au(CN)}_2]^- + 4\text{OH}^-$, recovered by zinc displacement. **Mond Process:** Nickel purified via volatile $[\text{Ni(CO)}_4]$.
- **3. Biological Role: Chlorophyll:** Coordination complex of Mg^{2+} (photosynthesis). **Hemoglobin:** Complex of Fe^{2+} with protoporphyrin IX (oxygen carrier). **Vitamin B_{12}:** Complex of Co^{3+} (corrin ring).
- **4. Medicinal Chemistry: Cisplatin (cis-$[\text{Pt(NH}_3)_2\text{Cl}_2]$):** Potent anti-cancer agent. **Chelation Therapy:** Ca-EDTA^{2-} for lead poisoning; D-penicillamine for Wilson disease.

2. JEE Advanced Core Master Reasoning Bank

Exam Diagnostic Question	Core Theoretical Principle	Precise Exam Explanation
Why is $[Fe(H_2O)_6]^{3+}$ colored while $[Mn(H_2O)_6]^{2+}$ is very pale?	Laporte & Spin Selection	Mn^{2+} is high-spin d^5; any $d\text{-}d$ transition requires spin flip (spin-forbidden). Fe^{3+} has charge transfer enhancement.
Why is $[CoF_6]^{3-}$ paramagnetic while $[Co(NH_3)_6]^{3+}$ is diamagnetic?	Splitting vs Pairing	F^- is weak field ($\Delta_o < P \implies t_{2g}^4 e_g^2$, 4 unp). NH_3 is strong field ($\Delta_o > P \implies t_{2g}^6 e_g^0$, 0 unp).
Why do tetrahedral complexes never form low-spin states?	Magnitude of Δ_t	$\Delta_t = \frac{4}{9}\Delta_o$. Electrostatic field is weak, so $\Delta_t < P$ always holds.
Why does $[Ni(dmg)_2]$ precipitate selectively?	Hydrogen Bonding	Planar arrangement with 2 symmetrical O-H···O bridges forms a neutral, highly insoluble chelate lattice.

Table 18.1 — Authoritative JEE Advanced reasoning traps and diagnostic answers.

TRAP

FINAL EXAM WARNING: In metal carbonyls, backbonding reduces the C-O bond order and increases the M-C bond order! Always check whether the question asks about M-C or C-O bond length before answering!